

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN

COURSE CODE: CSA1105

COURSE OUTCOME COVERED

***CO1:** Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr Ramamoorthy

QUESTIONS: 6

Total Marks:30

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I Darshinie Karani.VM (!92520033)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Short Answer Test

1. Suppose a software system is developed for a **Smart Hotel Management System** where guests can book rooms, cancel reservations, and make online payments.

(a) Explain the following object-oriented concepts:

- Class
- Object
- Inheritance
- Encapsulation
- Dynamic Binding

(b) Describe the following object relationships:

- Association
- Dependency
- Composition

(c) Explain how these concepts improve software maintainability and code reusability in a hotel management system.

(5 Marks)

2. Consider developing a **Smart Education Management System**.

(a) Explain the following object-oriented design principles:

- Abstraction
- Encapsulation
- Information Hiding

(b) Describe the following SOLID principles:

- Single Responsibility Principle (SRP)
- Dependency Inversion Principle (DIP)

(c) Explain how these principles improve software scalability and maintainability.

(5 Marks)

3. Suppose a software company is developing a **Smart Pharmacy Management System**.

(a) Explain the phases of the Software Development Life Cycle (SDLC):

- Feasibility Study
- Requirement Analysis
- System Design
- Implementation
- Testing
- Maintenance

(b) Describe the significance of each SDLC phase.

(c) Explain how SDLC helps reduce project risks and improve software quality.

(5 Marks)

4. Consider a project for developing a **Smart E-Commerce System**, where customer requirements frequently change due to market demands.

(a) Explain the following SDLC models:

- Waterfall Model
- V-Model
- Incremental Model
- Agile Model

(b) Compare the Waterfall Model and Incremental Model.

(c) Recommend the most suitable SDLC model for this project with justification.

(5 Marks)

5. Suppose a software analyst is designing a **Smart Hospital Information System**.

(a) Explain the role of UML in object-oriented software design.

(b) Describe the following UML diagrams:

- Use Case Diagram
- Sequence Diagram
- Communication Diagram
- Activity Diagram

(c) Explain how UML diagrams support software analysis, design, and documentation.

(5 Marks)

6. Consider developing a **Smart Railway Reservation System**.

(a) Explain the concept of object-oriented software design.

(b) Compare the structured software development approach with the object-oriented software development approach.

(c) Explain how object-oriented design improves software flexibility, scalability, and reusability.

(5 Marks)

RUBRICS FOR CALCULATION

SHORT ANSWER TEST

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	20%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Answer

1. a)

Class: class is used to create objects.

Object: it is derived from class.

Inheritance: One class gets all properties from another class.

Encapsulation: Keeping data and methods together and protects the data.

b)

Association: two objects are connected together.

Dependency: A object is dependent on another object.

Composition: The object cannot exist without another.

c)

- It is easy to use and maintain
- Reduces errors
- Code can be reused

2. a)

Abstraction: Shows only important datas and hides the other

Encapsulation: Keeping data and methods together and protects the data.

Information hiding: Prevents direct access from using the other data

b)

SRP: one class should do one job.

DIP: Depends on other interfaces, not directly on the class.

c)

- Easy to add new features.
- Easy to maintain.
- Better software performance.

3. a)

Feasibility Study : Checks if the project is possible

Requirement Analysis : Collects user needs

System Design : Plans how the software will work

Implementation: Writes the program code

Testing : Finds and fixes errors

Maintenance: Updates and improves the software after release

b)

- Feasibility – Saves time and money.
- Requirements – Understands user needs.
- Design – Gives a clear plan.
- Implementation – Builds the software.
- Testing – Removes bugs.
- Maintenance – Keeps software updated.

c)

- Reduces project risks.
- Improves software quality.
- Helps complete the project successfully.

4. a)

- Waterfall Model:
Each phase is completed one after another.
- V-Model:
Testing is done along with every development stage.
- Incremental Model:
Software is developed in small parts.
- Agile Model:
Software is developed in short cycles with regular changes.

b)

Waterfall	Incremental
One step at a time	Developed in small parts
Changes are difficult	Changes are easy
Final product comes at the end	Product is delivered step by step

c)

Agile Model, because customer requirements change frequently.

5. a)

Role of UML: UML helps to design the program and understands the relationship between them and then we construct the software before coding.

b)

- Use Case Diagram:
Shows users and system functions.
- Sequence Diagram:
Shows the order of messages between objects.
- Communication Diagram:
Shows how objects communicate.
- Activity Diagram:
Shows the workflow of activities.

c)

- Makes software easy to understand.
- Helps in design.
- Improves documentation.
- Reduces mistakes.

6. a) Object-Oriented Software Design :

Software is designed using classes and objects.

b)

Structured	Object oriented
Uses functions	Uses classes and objects
Less reusable	Highly reusable
Difficult to maintain	Easy to maintain

c)

- Flexible to changes.
- Easy to expand.
- Reuses existing code.
- Easy to maintain.